
Basic Risk Learning: Key Concepts in Financial Markets

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1. Introduction to Risk in Financial Markets

- **What is Risk?**
 - Risk is the possibility of a negative outcome, or the variability of returns, in an investment or financial decision.
 - Types of risk: Systematic risk, unsystematic risk, market risk, credit risk, etc.
 - The role of risk in financial decision-making: Risk and return are two fundamental elements in finance; the higher the risk, the higher the potential return (or loss).
 - **Risk and Uncertainty**
 - Risk involves the possibility of a defined outcome, whereas uncertainty involves a situation with unknown outcomes.
 - Measuring risk: Volatility, standard deviation, beta, Value at Risk (VaR), and other metrics.
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2. The No Arbitrage Principle

- **What is Arbitrage?**
 - Arbitrage refers to the practice of exploiting price discrepancies in different markets for the same asset, earning a risk-free profit.
 - Example: If an asset is priced at \$100 in one market and \$105 in another, an arbitrageur could buy low and sell high, making an instant profit.

- **The No Arbitrage Principle**

- The **No Arbitrage Principle** states that in efficient markets, arbitrage opportunities do not exist or are quickly exploited by traders, ensuring that prices remain fair and aligned.
- **Mathematical Formulation:**
 - The principle asserts that the price of a derivative should not deviate significantly from its theoretical price (derived from the spot market or underlying asset), otherwise, arbitrageurs will correct the discrepancy.

- **Importance of the No Arbitrage Principle in Pricing**

- For forward, futures, and option markets to be efficient, they must adhere to the no arbitrage condition.
 - **Example:** If forward prices deviate from their theoretical value, traders will exploit the differences, bringing prices back into alignment.
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3. Pricing of Forward Contracts

- **What is a Forward Contract?**

- A forward contract is a financial agreement to buy or sell an asset at a predetermined price on a specified future date.
- Unlike futures contracts, forwards are customizable and traded over-the-counter (OTC), not on exchanges.

- **Forward Pricing Formula**

- **Formula:** $F_0 = S_0 \times (1 + r)^t$

- Where:

- F_0 is the forward price at time t ,
 - S_0 is the spot price at the current time,
 - r is the risk-free interest rate,
 - t is the time to maturity in years.
- Example: If the spot price of a commodity is \$100, the risk-free rate is 5%, and the time to maturity is 1 year, the forward price will be calculated as:

- $F_0 = 100 \times (1 + 0.05)^1 = 105.$
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- **Arbitrage and Forward Pricing**

- Arbitrage ensures the forward price is consistent with the no arbitrage principle.
 - Any deviation from the formula would prompt arbitrage trading, ensuring the forward price returns to equilibrium.
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4. Pricing of Futures Contracts

- **What is a Futures Contract?**

- Futures are standardized contracts traded on exchanges that obligate the buyer to purchase (or the seller to sell) an asset at a specified future date and price.
- Unlike forwards, futures are marked to market daily, and require daily margin payments.

- **Futures Pricing Formula**

- **Formula:** $F_0 = S_0 \times (1 + r)^t$
 - This is the same as the forward pricing formula, but for futures, the daily settlement can impact the pricing.
- Example: If the spot price is \$200, the risk-free rate is 6%, and the time to maturity is 0.5 years, the futures price is:
 - $F_0 = 200 \times (1 + 0.06)^{0.5} \approx 203.00$.

- **Difference Between Futures and Forward Contracts**

- Futures are standardized and traded on exchanges, whereas forwards are customized and traded OTC.
- Futures have daily margin calls and are marked to market, unlike forwards that settle at maturity.

- **Arbitrage in Futures Markets**

- Just like forward contracts, futures prices must align with their theoretical price. Arbitrage opportunities will arise if there's a mismatch.

- Now, How to check if the price of Future contract is fairly valued?. Before that lets look at the following example:

Illustrations:

Cash and carry arbitrage

The following data is available on stock A as on August 1, 2008.

Cash market price	Rs. 1500
December Futures	Rs. 1520
Contract multiplier for stock	100 shares
Assume an implied cost of carry of 8% p.a. i.e. 0.75% per month.	

Theoretically/ fair price of December futures is 1504.69 ($= 1500 * e^{0.0075*5/12}$). Going by the theoretical price, we may say that December futures on stock A are overvalued. To take advantage of the mispricing, an arbitrageur may buy 100 shares of stock A and sell 1 futures contract on that at given prices. This would result in the arbitrage profit of Rs. 1531 ($= 100 \times 15.31$), which is the difference between actual and fair prices for 100 shares. Position of the arbitrageur in various scenarios of stock price would be as follows:

If you are using excel you can apply formula: $=1500*EXP(0.0075*5/12)$

Case I: Stock rises to Rs. 1550

Profit on underlying = $(1550 - 1500) \times 100 = \text{Rs. } 5000$

Loss on futures = $(1550 - 1520) \times 100 = \text{Rs. } 3000$

Gain on Arbitrage = Rs. 2,000

Cost of Arbitrage in terms of financing (Rs. 4.69 for 100 shares) = Rs. 469

Net gain out of arbitrage = $(2000 - 469) = \text{Rs. } 1531$

Case II: Stock falls to Rs.1480

Loss on underlying = $(1500 - 1480) \times 100 = \text{Rs. } 2000$

Profit on futures = $(1520 - 1480) \times 100 = \text{Rs. } 4000$

Gain on Arbitrage = Rs. 2000

Cost of Arbitrage in terms of financing (Rs. 4.69 for 100 shares) = Rs. 469

Net gain out of arbitrage = $(2000 - 469) = \text{Rs. } 1531$

Lets Look at the live example on NSE:

Today's Date is 20 Jan 2024 and Nifty Spot Price is 23,220



Search by company name, symbol or keyword

English



23,220.00
16.80 (0.07%)
20-Jan-2025 09:30

Market Capitalization
₹ Lac Crs 425.94 | Tn \$ 4.92
17-Jan-2025

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NIFTY

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DERIVATIVES

As on 20-Jan-2025 09:29:44 IST (All price values in Rs)

All Contracts Option Chain All Historical Data Glossary

INSTRUMENT TYPE	EXPIRY DATE	OPTION	STRIKE	OPEN	HIGH	LOW	CLOSE	PREV. CLOSE	LAST	CHNG	%CHNG	VOLUME (Contracts)	VALUE (₹ Lakhs)
Index Futures	30-Jan-2025	-	-	23,339.95	23,340.00	23,240.00	-	23,267.20	23,255.95	-11.25	-0.05	16,522	96,160.85
Index Futures	27-Feb-2025	-	-	23,457.80	23,473.50	23,375.00	-	23,402.60	23,394.30	-8.30	-0.04	463	8,132.37
Index Futures	27-Mar-2025	-	-	23,602.40	23,615.30	23,530.00	-	23,551.45	23,545.40	-6.05	-0.03	152	2,686.60

Assuming the Cost of Carry at 8% (Remember the 10y Gsec yield is ~7%)

Now to check if 27, Feb 2025 price is fair or not you can apply this formula in excel =
 $=23,220 * \text{EXP}(8\% * 38/365)$ fair price as per calculation will be 23,414.2

Now if you compare fair price as per calculation and quoted price on exchange which is 23,394.30 there is hardly any difference of ~20. Which is just 0.08% hence, there is no arbitrage opportunity.

5. Properties of Options

• What are Options?

- An option is a financial derivative that gives the holder the right, but not the obligation, to buy or sell an asset at a predetermined price within a specified time period.
- There are two main types of options:
 - **Call Options:** The right to buy an asset.
 - **Put Options:** The right to sell an asset.

• Key Properties of Options

- **Strike Price:** The price at which the underlying asset can be bought or sold.
- **Premium:** The cost of the option.
- **Expiration Date:** The date by which the option must be exercised.
- **Intrinsic Value:** The value of the option if it were exercised today.
- **Time Value:** The additional value based on the time remaining until expiration.

• Option Pricing (The Black-Scholes Model)

- **Formula:** The Black-Scholes model provides a way to calculate the theoretical price of European call and put options.
- The formula is:

$$C = S_0N(d_1) - Xe^{-rt}N(d_2)$$

Where:

- C is the call option price,
 - S_0 is the current stock price,
 - X is the strike price,
 - t is the time to expiration,
 - r is the risk-free interest rate,
 - $N(d_1)$ and $N(d_2)$ are the cumulative distribution functions of the standard normal distribution.
- **Properties of Option Prices**
 - **Time Decay (Theta):** The value of options decreases as expiration approaches.
 - **Volatility (Vega):** The value of options increases with volatility of the underlying asset.
 - **Delta:** The sensitivity of the option price to changes in the underlying asset price.
 - **Gamma:** The rate of change of delta in response to changes in the asset price.

Note: We will learn detailed Option Pricing (The Black-Scholes Model) in Unit 8 – Modern Financial Theory.